

State Economic Freedom and Firm Size as Moderators of Paid Family Leave's Effect on Firm Risk-Taking

1. Introduction

Whether and how public social-insurance policy reshapes corporate behavior sits at the intersection of public economics and strategic management. Paid Family Leave (PFL) is a salient case: a state-administered, payroll-funded program guaranteeing partial wage replacement to workers who take leave for a new child or an ill relative. Lin (2025) shows that the staggered adoption of PFL across U.S. states increased firm strategic risk-taking, attributing the effect to managerial cognition: executives interpret the mandate as a threat and respond with compensatory, risk-seeking behavior. Lin also reports that the effect is weaker for small and medium enterprises.

That finding establishes an average effect and one size contingency. This study takes up a further question, one of conditions. If the response is behavioral and channeled through managerial choice, its magnitude should depend on how much latitude managers have to act. We therefore add state economic freedom as a moderator, and our lead theoretical move is to connect it to Lin's firm-size moderator: managerial-discretion theory treats both the external task environment (economic freedom) and the internal organization (firm size) as sources of the same underlying latitude, so the two are connected as external and internal conditions rather than independent additions, with a testable interaction.

The study makes two contributions. The first is a theoretical framework: it connects two previously separate moderators under managerial discretion and shows how their interaction provides a directional way to compare two competing accounts of how firm size and the institutional environment jointly shape the policy response. We offer this as a promising lens to be tested, not as a demonstrated result. The second is methodological: it re-estimates the size moderation with staggered-robust estimators (Lin used two-way fixed effects, which can be biased under staggered timing when treatment effects are heterogeneous) and conducts inference appropriate to a small number of treated states. Empirically, the analysis is an honest, power-limited pilot: what the data support most clearly is that firm size matters, while the economic-freedom moderation and the

firm-size-by-freedom connection are directional rather than significant, and we report them as such.

2. Theoretical framework

2.1 Lin's foundation

The study builds on Lin (2025), whose thesis is that PFL adoption causes firms to increase strategic risk-taking, with managerial cognition as the operative mechanism. The argument rests on four established theories we accept and extend. Threat-opportunity categorization (Dutton and Jackson, 1987) holds that managers categorize strategic issues as threats or opportunities, and threats trigger defensive, often risk-seeking behavior. Upper-echelons theory (Hambrick and Mason, 1984; Hambrick, 2007) holds that firm-level choices reflect the cognition and values of top executives. The behavioral agency model (Wiseman and Gomez-Mejia, 1998), drawing on prospect theory (Kahneman and Tversky, 1979), holds that managers below a reference point become risk-seeking. Agency theory (Jensen and Meckling, 1976) supplies the misalignment that turns a threat-framed shock into immediate strategic responses. Lin also examines firm size as a moderator and excludes Rhode Island from the main analysis for failing a pre-trend diagnostic; we inherit both choices.

Upper-echelons theory is the natural home of this extension. Its core claim is that firm outcomes reflect the cognition, experience, and values of the dominant coalition of top executives, so the threat response Lin documents is, concretely, the response of CEOs and their top teams. Crucially, upper-echelons effects are not constant: Hambrick (2007) stresses that executives' dispositions translate into firm outcomes most strongly where they enjoy discretion and are muted where they do not. The natural next question is therefore how much latitude these executives have to act on their threat interpretations, which is the subject of managerial-discretion theory and the hinge of this extension.

2.2 Managerial discretion (the lead theory)

Managerial-discretion theory is traced to Hambrick and Finkelstein (1987), which reconciles population ecology (the environment selects; managers scarcely matter) with strategic choice (managers shape firms) by introducing discretion: the latitude of action available to decision-makers. Where discretion is high, managerial and environmental forces translate strongly into strategic outcomes; where it is low, outcomes are constrained regardless of intent. Discretion is determined by three forces: the task environment, the internal organization, and managerial

characteristics (Wangrow, Schepker, and Barker, 2015). Two of these motivate our moderators: economic freedom captures task-environment latitude, and firm size captures internal-organization latitude. Discretion, however, is not only freedom from constraint; acting on it also requires resources, and firm size proxies both, since larger firms may carry more bureaucracy but also more financial slack while SMEs have less of each. In upper-echelons terms, discretion is the gate between an executive's threat interpretation and the firm's realized risk-taking: the same threat framing produces a larger strategic response where managers have more room, and more resources, to act. The unifying prediction is that the PFL effect is amplified where discretion is greater, with the size moderation the place where organizational flexibility and resource capacity can pull in opposite directions (Section 2.4).

2.3 Economic freedom as task-environment discretion

The institutional environment systematically shapes corporate risk-taking. John, Litov, and Yeung (2008) show that where institutions protect property and constrain private benefits, managers pursue riskier, higher-return projects; their specific institution is investor protection, which EFNA does not measure, so we use the work for the general principle and obtain the construct match elsewhere. Economic freedom (in the market-process tradition, the capacity to respond to opportunity: light regulation, limited fiscal burden, and labor-market flexibility, with secure property and voluntary exchange in the broader tradition) is theorized to enlarge that capacity (Hayek, 1945; Kirzner, 1973), within New Institutional Economics (North, 1990; Gwartney, Lawson, and Block, 1996). The subnational EFNA index used here captures the fiscal and labor-market dimensions specifically rather than property rights (Section 4.4). The evidence that economic freedom shapes real firm behavior is substantial: the literature surveyed by Hall and Lawson (2014), spanning hundreds of empirical studies, finds economic freedom robustly associated with entrepreneurship, investment, and growth. At the U.S. state level specifically, Bjørnskov and Foss (2008) link the Fraser index to entrepreneurial activity, and Lucas and Boudreaux (2019) show that state economic freedom moderates the effect of federal regulation on firms and that this moderation varies by firm characteristics, which is the closest precedent both for using state economic freedom as a firm-level moderator and for expecting its strength to depend on firm size. Where regulatory and fiscal friction is low, the additional risk-taking PFL induces passes through more fully into investment. This argument yields the economic-freedom moderation hypothesis (H1).

2.4 The connection between economic freedom and firm size

Firm size already moderates the PFL effect in Lin (2025). The contribution here is not to re-introduce it but to connect it to economic freedom: both are sources of managerial discretion, economic freedom external and firm size internal, so the two are connected as external and internal conditions shaping discretion rather than independent additions. Crucially, the supporting theories split into two camps that make opposite predictions for the interaction, which makes the interaction a way to compare the two accounts rather than a control.

Camp A, complementarity (predicts a positive interaction: economic freedom matters more for small firms). Several literatures imply small firms are more exposed to, and more dependent on, the institutional environment. Regulatory and compliance costs are substantially fixed, so their per-employee burden tends to fall with size and small firms may bear a disproportionate share (Crain and Crain, 2010, a Small Business Administration report whose aggregate cost estimates have been contested, so it is treated as suggestive rather than as sole support), making economic freedom more consequential for them. The behavioral theory of the firm and the resource-based view note that large firms hold slack and resources that buffer them from the environment while small firms do not (Cyert and March, 1963; Bourgeois, 1981; Penrose, 1959). The liability of smallness (Aldrich and Auster, 1986) makes small firms more vulnerable to, and dependent on, a favorable environment, and transaction-cost reasoning (Williamson, 1985; North, 1990) implies small firms rely more on the market environment than on internal hierarchy. All predict that the economic-freedom moderation is stronger among SMEs, a positive triple interaction.

Camp B, financial constraints (predicts a negative interaction: small firms are too constrained to respond at all). Small firms face binding capital-market frictions (Fazzari, Hubbard, and Petersen, 1988; Hadlock and Pierce, 2010). If they cannot finance additional risk-taking, they respond little to PFL regardless of the institutional environment, so the PFL effect, and any amplification of it by economic freedom, concentrates in large firms. This predicts a negative triple interaction.

Put simply, the two camps disagree about what binds a small firm: economic freedom supplies latitude (relief from the regulatory, tax, and labor-market friction that falls disproportionately on small firms) but not capacity (the capital needed to finance a risk-taking response). Camp A is the case where an SME's binding constraint is friction, so freedom lets it act; Camp B is the case where

the constraint is financing, which freedom cannot relax. The two accounts imply opposite signs for the triple interaction, so its sign indicates which force appears dominant in this setting; the accounts need not be mutually exclusive, since small firms can be both more exposed to regulation and more financially constrained. That contrast, not the existence of either moderator alone, is the empirical payoff of connecting them (H3).

2.5 Integration

The integrated account is a moderated process. Threat-framing initiates a heightened willingness to take risk (Lin, 2025). Whether that willingness becomes action depends on discretion, whose external source (economic freedom) and internal source (firm size) jointly condition the effect and interact, with firm size governing exposure to the institutional environment. The remainder of the paper states these claims as hypotheses, describes the data and measures, defines the empirical strategy, and reports an empirical pilot.

3. Research questions and hypothesis development

Primary research question. Do state economic freedom and firm size jointly moderate the effect of PFL on firm strategic risk-taking, and are they connected as two sources of managerial discretion?

Each hypothesis is derived from a building block in Section 2.

Baseline (main effect, following Lin). From threat-opportunity categorization (Dutton and Jackson, 1987) and the behavioral agency model (Wiseman and Gomez-Mejia, 1998), PFL is framed as a threat and raises firm risk-taking. This is Lin's (2025) result, taken as the premise the moderation builds on.

H1 (economic-freedom moderation). From the task-environment component of managerial discretion (Hambrick and Finkelstein, 1987) and the institutions-and-risk-taking principle (John, Litov, and Yeung, 2008; Bjørnskov and Foss, 2008), the effect of PFL on firm risk-taking is stronger in states with higher baseline economic freedom, because fiscal and labor-market freedom provide external latitude for firms to act (a positive interaction).

Rival to H1 (substitution). If high-freedom states already provide ample latitude, the marginal PFL impulse is smaller there, implying a negative interaction; the sign adjudicates.

H2 (firm-size moderation, following Lin). From the internal-organization component of managerial discretion, the effect of PFL on firm risk-taking varies with firm size, because firm size captures internal capacity and latitude such as organizational slack and bureaucracy. Lin (2025) finds the effect weaker for SMEs; whether that reflects complementarity or financial constraints is the question H3 addresses.

H3 (the economic-freedom by firm-size connection; lead contribution). The interaction between economic freedom and SME status provides a directional way to compare two accounts. A positive triple interaction supports complementarity: economic freedom matters more for exposed SMEs. A negative triple interaction supports financial constraints: SMEs are too constrained to convert external freedom into additional risk-taking. The sign indicates which account the data lean toward, though with three treated states the comparison is low-powered.

4. Data and measures

4.1 Sources

The unit of analysis is the firm-year. The firm panel is Lin's Compustat manufacturing sample with executive characteristics, merged with state-level moderators and the SME indicator.

Table 1. Data sources.

Variable	Source	Coverage
Firm financials and risk-taking	Compustat (via Lin, 2025)	2002-2019
Executive characteristics	Execucomp	2002-2019
PFL adoption years	State legislation records	2004-2023
Economic freedom (moderator)	Fraser Institute, EFNA subnational	1981-2023
SME status (moderator)	SBA / NAICS size standards	2002-2019
State controls	BEA, BLS	2002-2019

Note: the main analysis uses fiscal years 2002 to 2019 (pre-COVID); the panel extends to 2022, and those years enter only the post-COVID robustness check (Section 6.5).

4.2 Treatment

Treatment is dated by the year PFL benefits become effective (payable), not the enactment year: California (2004), New Jersey (2009), Rhode Island (2014), and New York (2018). Following Lin (2025), Rhode Island is excluded from the main analysis because it fails the pre-trend diagnostic, leaving three treated cohorts (California, New Jersey, New York). We reproduce that diagnostic rather than take it on faith: plotting Rhode Island's mean risk-taking against the never-treated control mean (Appendix B, Figure A1) shows the two series tracking each other through about 2005 and Rhode Island then rising well before its 2014 effective date, a clear parallel-trends violation that confirms Lin's reasoning. Because Rhode Island is also a very small cohort (4 firms, 51 firm-years), the plot is descriptive rather than a powered test; the main results are in any case robust to including Rhode Island (Section 6.5). Later adopters whose benefits became payable after the estimation window (Washington 2020, Massachusetts 2021, Connecticut 2022, Oregon 2023, and the District of Columbia 2020) are never- or not-yet-treated within the pre-COVID window and serve as controls; the panel contains two firms headquartered in the District of Columbia, which are controls because D.C. benefits postdate the window. The California cohort dominates the treated firm-years (205 of 277 treated firms and 1,872 of 2,589 treated firm-years), which is relevant to the power of the staggered estimators.

Table 2. Treated cohorts: timing, baseline economic freedom, and size.

State	EFNA baseline	Implied PFL effect
NY	4.40	0.105
RI	4.65	0.127
CA	4.80	0.142
NJ	6.33	0.282

Note: national median baseline economic freedom is 6.27 across 50 states; New Jersey is the only treated cohort above it. Rhode Island is excluded for a pre-trend violation (Section 4.2).

4.3 Outcome

Firm strategic risk-taking (`factor_score`), a standardized factor of R&D intensity, capital expenditure, and long-term leverage (Devers et al., 2008; Martin, Gomez-Mejia, and Wiseman,

2013); the same outcome as Lin (2025), so coefficients read in standard-deviation units. Because R&D (Compustat xrd) is frequently left blank rather than zero for firms that do not separately report it, the panel follows Lin in keeping only firm-years with non-missing, non-negative R&D (xrd at least zero) before the factor is built; R&D is therefore fully populated in the analysis sample (zero percent missing, with 64 firm-years reporting exactly zero), and capital expenditure, leverage, and assets are likewise complete. This is a sample-selection choice that conditions on R&D-reporting firms, which in this manufacturing panel are the large majority; a natural robustness alternative sets missing R&D to zero (the common finance convention) and re-includes those firm-years.

4.4 Moderators

Economic freedom. The EFNA subnational overall score, which captures state-level fiscal and labor-market institutional freedom through three areas (government spending, taxation, and labor-market regulation). Because economic freedom is a property of the state rather than the individual firm, it enters the analysis at the firm level the way any contextual moderator does: each firm is assigned its headquarters state's score, so it varies across firms only through headquarters location and is fixed within firm over the baseline window. This is the same firm-level use of a state institutional measure as in Lin (2025) for state controls and in Lucas and Boudreaux (2019); the question it poses is not whether economic freedom is itself a firm characteristic, but whether the state environment a firm sits in conditions how that firm responds to PFL. The subnational index does not include the legal-system/property-rights and trade areas of Fraser's all-government cross-country index, so the moderator should be read as fiscal-and-labor freedom rather than property-rights protection. It is averaged over a pre-treatment window (2000 to 2003) and mean-centered, so its main effect is absorbed by firm fixed effects while its interaction is identified. Because California enacted PFL in 2002 (benefits payable 2004), this window is pre-benefit but not fully pre-enactment for California; we therefore report robustness to strictly pre-enactment baselines (1998 to 2001, and 2000 to 2001) and to a leave-California-out check.

Firm size (SME). The SBA size-standard indicator, which classifies a firm as an SME if its employment is at or below its industry's threshold. These thresholds are industry-specific (twenty distinct employee cutoffs from 500 to 1,500 across the NAICS manufacturing sub-sectors in this sample), so SME status measures smallness relative to a firm's own industry rather than an absolute

headcount; firms in industries with receipts-based rather than employee-based standards lack a classification and are excluded from the size-stratified analyses. Because a firm's status can change as it grows, we fix it at the firm's earliest non-missing value, paralleling the economic-freedom moderator and avoiding conditioning on a post-treatment outcome. For treated firms that enter Compustat after their state's adoption, this earliest value can itself be post-treatment; such firms are flagged and a robustness check restricts treated firms to those with a genuine pre-treatment size observation. Because the SBA revises its standards periodically, a further check uses contemporaneous-year thresholds rather than a single fixed vintage.

4.5 Sample restrictions

The sample is the pre-COVID window (fiscal years through 2019) and is restricted to firms with a stable headquarters state across the window, so that treatment timing, cohort, and the baseline moderators are well defined. From a raw 10,468 firm-years (913 firms), restricting to the pre-2019 window and to stable-headquarters firms drops 32 relocating firms, and excluding Rhode Island (Section 4.2) leaves 8,761 firm-years on 856 firms: 277 treated firms (2,589 treated firm-years, of which California supplies 1,872) and 579 control firms. Across all firms, 262 are baseline SMEs and 537 are larger firms, with the remaining 57 in industries that lack an employee-based size standard. Standard errors are clustered by state (Bertrand, Duflo, and Mullainathan, 2004); with only three treated states the design is power-limited, a caveat that applies throughout.

5. Methodology

5.1 Causal parameter of interest

The design targets the average treatment effect on the treated (ATT): the expected effect of PFL on risk-taking among treated firm-years, identified under conditional parallel trends and no anticipation, and estimated free of staggered-timing bias by Callaway-Sant'Anna (2021). This is the parameter the moderation analysis conditions on; the moderators then provide evidence on how that effect varies with the external (economic freedom) and internal (firm size) sources of managerial discretion.

5.2 Estimators

The benchmark specification is a two-way fixed-effects difference-in-differences. The baseline, which follows Lin (2025), is:

$$RiskTaking_{i,t} = \beta PFL_{s,t} + X'_{i,t}\gamma + \alpha_i + \delta_t + \varepsilon_{i,t}$$

and the full triple-difference, the lead specification, is:

$$RiskTaking_{i,t} = \beta_1 PFL_{s,t} + \beta_2 (PFL_{s,t} \cdot EconFree_s) + \beta_3 (PFL_{s,t} \cdot SME_i) + \beta_4 (PFL_{s,t} \cdot EconFree_s \cdot SME_i) + X'_{i,t}\gamma + \alpha_i + \delta_t + \varepsilon_{i,t}$$

Here *RiskTaking* is the firm risk-taking factor for firm *i* in year *t*; PFL equals one once firm *i*'s state *s* has paid family leave benefits payable in year *t*; *EconFree* is the state's baseline (2000 to 2003), mean-centered economic freedom; *SME* is the baseline size indicator; *X* is the vector of firm and state controls (firm size, CEO age and gender, industry concentration, sales-growth volatility, and state unemployment, population, and GDP), the same set used by Lin (2025); and α and δ are firm and year fixed effects, with standard errors clustered by state. In the baseline, β is the average PFL effect. In the triple-difference, because economic freedom is mean-centered and SME is zero for large firms, β_1 is the PFL effect for non-SME firms at mean economic freedom; β_2 is the economic-freedom moderation among non-SMEs (H1); β_3 is the SME difference in the PFL effect at mean economic freedom (H2); and β_4 is whether the economic-freedom moderation differs for SMEs (H3), whose sign points toward Camp A or Camp B. The implied PFL effect is $\beta_1 + \beta_2 \cdot EconFree$ for non-SME firms and $\beta_1 + \beta_3 + (\beta_2 + \beta_4) \cdot EconFree$ for SMEs. The time-invariant moderators have no standalone terms because the firm fixed effect absorbs them; only their interactions with PFL are identified. The specification menu in Section 6 reports A1 (β_2 only), A2 (β_3 only), B (β_2 and β_3), and C (all four).

The strategy proceeds in four layers. First, the two-way fixed-effects (TWFE) baseline above estimates the average PFL effect, absorbing time-invariant firm differences and common year shocks. Second, the triple-difference (DDD) adds PFL interacted with economic freedom, with SME status, and with both, so the moderation is read directly from the coefficients; because both moderators are time-invariant, the firm fixed effect absorbs their levels and only their interactions with PFL are identified. These two are benchmarks: under staggered adoption with heterogeneous effects, TWFE and DDD can be biased because already-treated firms are used as controls (Goodman-Bacon, 2021). Third, the Callaway-Sant'Anna (2021) estimator removes that bias by computing cohort-by-time ATTs that compare each treated cohort only with not-yet-treated firms, then aggregating them into a single combined (overall) ATT, which is the staggered-robust analogue of the TWFE main effect. Fourth, because Callaway-Sant'Anna has no native interaction

term, the moderation is recovered by stratification: the estimator is re-run within subgroups, high versus low baseline economic freedom and SME versus non-SME, and the subgroup ATTs are compared, which is the staggered-robust analogue of the interaction and upgrades Lin's TWFE-only size moderation. Both the combined ATT and the stratified ATTs are reported (Table 4). All Callaway-Sant'Anna estimates use the not-yet-treated as the comparison group, no anticipation, an unbalanced panel, and standard errors clustered by state; the stratified ATTs use doubly-robust estimation, the combined ATT a state-cluster bootstrap, and a simple aggregation reproduces the stratified pattern, confirming it is not an artifact of the doubly-robust adjustment. In the regression layer the moderators are reported four ways: economic freedom alone (A1), SME alone (A2), both additively (B, each net of the other), and both with their interaction (C, the connection).

5.3 Identification and inference

Identification rests on the parallel-trends assumption together with the variation in PFL adoption across states; it does not come from the staggering of adoption itself. Staggered timing is an econometric complication rather than a source of identifying variation: it biases two-way fixed effects when treatment effects are heterogeneous (Goodman-Bacon, 2021), which is precisely why the Callaway-Sant'Anna estimator is used. Beyond this, two threats dominate. First, the small number of treated clusters: the interactions are identified off three treated states, so standard errors clustered by state are imprecise and the estimates are best read as directional; we report the difference-in-differences coefficients and their clustered standard errors directly, supported by a leave-California-out check. Second, selection into adoption: treated states are not random, addressed by Lin's pre-trend analysis (which excludes Rhode Island), the never- and not-yet-treated controls, and the baseline (pre-treatment) construction of all moderators.

6. Results

Estimates are the TWFE/DDD models and the stratified Callaway-Sant'Anna, computed on the stable-headquarters, pre-2019 panel with Rhode Island excluded (three treated states). With only three treated states the coefficients are best read as directional; standard errors are clustered by state. We present each specification in turn, the main effect, then each moderator alone, then the two together, then their interaction, so the contribution of every regression is visible separately (Table 3), before the combined and stratified staggered-robust estimates (Table 4) and the robustness checks (Table 5).

6.1 Main effect and consistency with Lin

PFL raises firm risk-taking: the TWFE estimate is +0.184. We reproduce the direction of Lin's (2025) main finding under a closely related specification: because our sample and measurement differ from hers, a baseline rather than contemporaneous SME measure, stable-headquarters firms only, Rhode Island excluded, and a pre-COVID window, we match Lin in sign and direction rather than claim a strict replication. The staggered-robust Callaway-Sant'Anna results (Section 6.3) point the same way, with the effect concentrated in larger firms. Lin's firm-size moderation (her triple term treated $\times$ post $\times$ is_sme) is -0.101 and significant; our estimate carries the same sign and direction, larger in magnitude for the same measurement reasons, and the size moderation is the most stable empirical pattern in our data.

6.2 Economic-freedom moderation (H1)

The interaction between PFL and baseline economic freedom is positive, 0.092; it remains positive but roughly halves when California is dropped (about 0.05), so the leave-California check is only weakly supportive. The implied PFL effect rises with a state's baseline freedom, from about 0.11 in New York to 0.14 in California and 0.28 in New Jersey, the only treated state above the national median (Figure 1). The positive sign is consistent with H1. The interaction is statistically significant on its own (clustered $p = 0.034$) but loses significance once firm size is added ($p = 0.23$), so with three treated states it is directional support for H1 rather than a confirmed result.

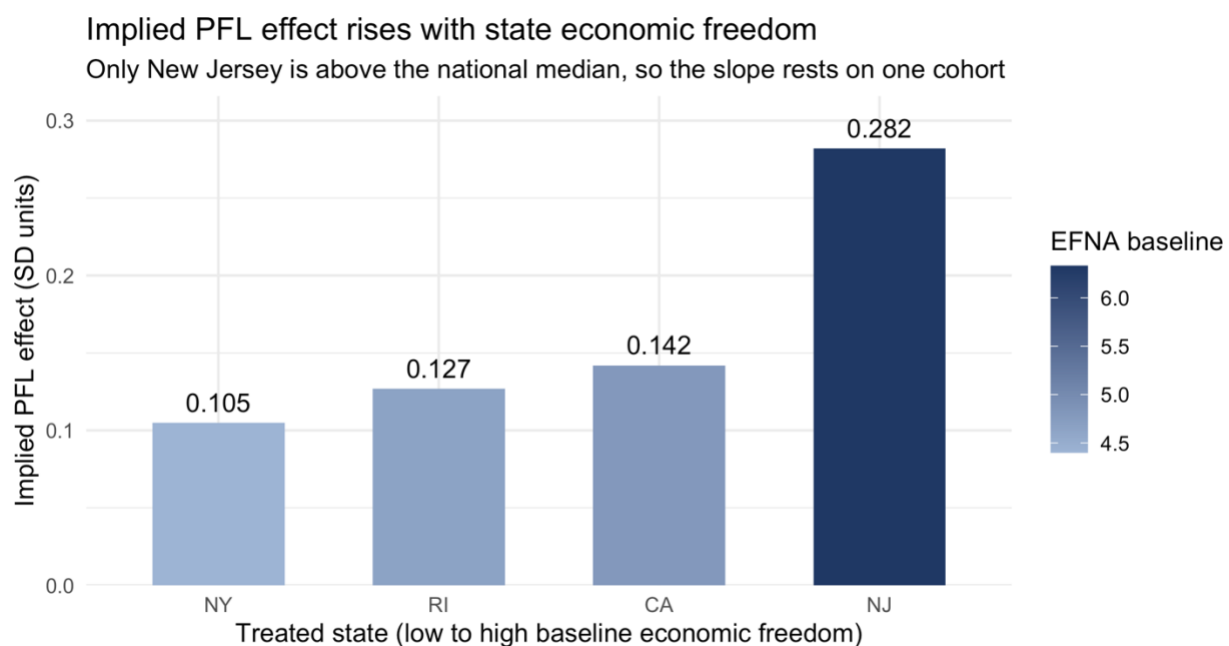


Figure 1. *Implied PFL effect by treated state (specification A1), ordered by baseline economic freedom. Only New Jersey lies above the national median, so the positive slope rests on a single cohort.*

6.3 Firm-size moderation (H2)

The firm-size moderation is the most stable empirical pattern in the draft. In the regression it is large and negative (about -0.4 across specifications). The staggered-robust Callaway-Sant'Anna stratification makes it concrete: the PFL effect is +0.237 for large (non-SME) firms and -0.070 for SMEs (Figure 2). In this sample, the PFL risk-taking effect appears concentrated among larger firms; SMEs barely respond. This survives the staggered-robust estimator and is the strongest finding in the study, and is our upgrade over Lin's TWFE-only estimate. A combined (pooled) Callaway-Sant'Anna ATT across all firms is positive (about +0.17) but, by averaging these opposite-sized groups, conceals the split, which is why the stratified estimates rather than a single combined number are the right summary here (Table 4).

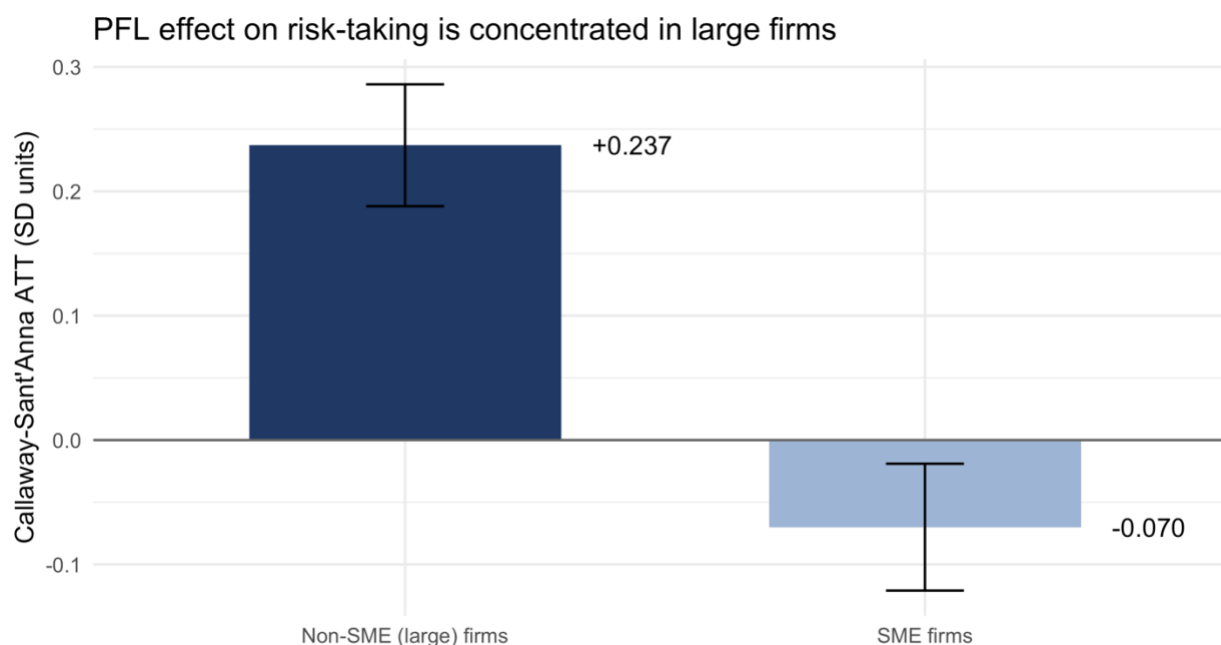


Figure 2. *Staggered-robust Callaway-Sant'Anna ATT by firm size. The PFL risk-taking effect is concentrated in large (non-SME) firms and is essentially zero for SMEs; whiskers are bootstrap standard errors.*

6.4 The connection (H3)

The triple interaction of PFL, economic freedom, and SME status is negative, -0.125, and the SME interaction it sits on is negative, so it leans (imprecisely) toward Camp B (financial constraints) rather than Camp A (complementarity): economic freedom does not appear to matter more for SMEs, and if anything its moderation weakens for them. This is consistent with the Section 6.3

pattern that SMEs barely respond to PFL at all, leaving little for economic freedom to amplify. The estimate is imprecise (a three-treated-state design split further by size is the least powered specification, with a clustered p-value of 0.163), so it is a directional result that is more consistent with one account than the other rather than a precise magnitude.

Table 3. Specification menu (TWFE/DDD coefficients; Rhode Island excluded). Standard errors clustered by state; p-values from the difference-in-differences models. With three treated states, read as directional.

Specification	Estimate	SE	p	Note
PFL (main effect)	0.184	0.066	0.008	replicates Lin
A1. PFL x EconFreedom	0.092	0.042	0.034	positive; weakens with SME
A2. PFL x SME	-0.405	0.096	0.000	effect smaller for SMEs
B. PFL x EconFreedom (with SME)	0.059	0.048	0.231	shrinks once SME held constant
B. PFL x SME (with EconFreedom)	-0.389	0.095	0.000	robust to adding EconFreedom
C. PFL x EconFreedom x SME	-0.125	0.088	0.164	negative; leans Camp B

The top row is in line with Lin's positive average effect (+0.184). Adding one moderator at a time, economic freedom (A1, +0.092) and firm size (A2, -0.405) are each significant on their own. The decisive comparison is row B, which includes both: the economic-freedom interaction shrinks to +0.059 and loses significance ($p = 0.23$), while the size interaction barely moves (-0.389) and remains statistically strong, so the apparent freedom effect was largely standing in for firm size. The triple interaction (C, -0.125) is negative, consistent with Camp B, but imprecise. Reading down the table, the size interaction remains large, stable, and statistically strong across A2, B, and C, though inference is still read cautiously because treated-state variation is limited; economic freedom, by contrast, is positive on its own but fragile once firm size is held constant.

Table 4. Callaway-Sant'Anna ATT, combined and stratified (staggered-robust), with bootstrap standard errors clustered by state. The two stratifications each decompose the combined ATT; each stratum identifies off one or two treated cohorts, so the splits are exploratory.

Stratum	ATT	Std. error
Combined (pooled, all firms)	0.086	0.058
High-freedom cohort (New Jersey)	0.100	0.032
Low-freedom cohorts (California, New York)	0.084	0.059
SME firms	-0.070	0.048
Non-SME (large) firms	0.237	0.048

Note: the combined row is the overall (pooled) ATT across all firms; the high/low and SME/non-SME blocks are two stratifications that each decompose it. Stratified standard errors are the doubly-robust multiplier-bootstrap errors from the Callaway-Sant'Anna estimator; the combined standard error is a state-cluster bootstrap of the same estimator. A simple aggregation reproduces the stratified pattern (non-SME about +0.22, SME about zero), confirming it is not an artifact of the doubly-robust adjustment.

The combined ATT is positive (+0.167), so on average PFL raises risk-taking under the staggered-robust estimator, in line with the TWFE main effect. The two stratifications then show where that average comes from. Splitting by firm size is decisive: the effect is large for non-SME firms (+0.237) and essentially zero, even slightly negative, for SMEs (-0.070), so the positive combined number is almost entirely a large-firm phenomenon. Splitting by economic freedom is not: the high-freedom (+0.100) and low-freedom (+0.084) cohorts differ little, echoing the weak regression interaction in Section 6.2. The table therefore tells the same story as the coefficients: firm size is the moderator that matters, while economic freedom barely moves the staggered-robust ATT.

6.5 Robustness

The headline patterns survive the design choices raised in review (Table 5). First, extending the window through 2022 to include the COVID years does not overturn the results: the main effect and the size moderation both strengthen and remain significant (the PFL effect is +0.25 and the SME gap -0.50), so the conclusions are not an artifact of the pre-COVID window, although the COVID period is an unusual shock and the post-2019 estimates should be read with that in mind.

Second, adding back the 32 firms that relocate headquarters leaves both terms significant and of the same sign (+0.16 and -0.31). Third, adding back Rhode Island, which is excluded from the main analysis for its pre-trend violation, barely moves the estimates (+0.17 and -0.38); the exclusion sharpens identification (Section 4.2) without driving the findings. The economic-freedom moderation and the triple interaction are likewise stable in sign across these checks (positive-but-weak and negative, respectively). Two assumptions cannot be fully discharged with these data and are flagged rather than resolved: the unusual nature of the COVID shock, and the assignment of treatment by headquarters state when firms operate across many states (Section 7).

Table 5. Robustness of the main effect and firm-size moderation (TWFE/DDD; standard errors clustered by state).

Sample / check	PFL (main)	PFL x SME
Baseline (≤ 2019 ; no RI, no movers)	+0.184***	-0.405***
+ Post-COVID years (through 2022)	+0.252***	-0.502***
+ Relocating firms (movers)	+0.163**	-0.314***
+ Rhode Island (excluded cohort)	+0.168**	-0.382***

*Note: *** $p < 0.01$, ** $p < 0.05$; standard errors clustered by state. Each row re-estimates the baseline (PFL) and size-interaction (PFL x SME) specifications on the indicated sample.*

7. Discussion and Conclusion

The main contribution is a framework, not a verdict. Firm size is Lin's (2025) moderator; economic freedom is the one this study adds; and managerial-discretion theory connects them as external and internal conditions shaping the same latitude. More than juxtaposing them, the framework shows how their interaction offers a directional way to compare a discretion-amplification account (Camp A) with a financial-constraints account (Camp B). With only three treated states we cannot run that test decisively; the available evidence leans, weakly, toward the financial-constraints reading, since SMEs barely respond to PFL and economic freedom therefore has little to amplify among them, but we present this as a direction for the framework to be tested against rather than as a finding. The methodological contribution is the staggered-robust re-estimation of the size

moderation, which Lin estimated only with two-way fixed effects, reported as directional given the handful of treated states.

Two empirical patterns deserve emphasis. The economic-freedom moderation is positive, with the largest implied effect in New Jersey, but with three treated states it is imprecise once firm size is accounted for; it is directional evidence, not a confirmed effect. The firm-size result is the most stable pattern and points against the complementarity prediction: the PFL effect is concentrated in large firms, consistent with small firms being too financially constrained to convert the threat response into financed investment. The result that treated states cluster in the lower half of the economic-freedom distribution is itself informative, consistent with the political-economy view that such mandates arise where the state already intervenes more.

Several limitations bound these conclusions. The moderation slopes are identified from only three treated states (Rhode Island excluded for a pre-trend violation, following Lin), so standard errors clustered by state are imprecise and the estimates are best read as directional; the novel results, economic freedom and the connection, are directional rather than significant, while the robust size moderation is inherited from Lin and re-estimated staggered-robustly here. The SME measure depends on the SBA size-standard construction and is restricted to industries with employee-based thresholds; the sample is manufacturing-only and pre-COVID; and treatment is assigned by headquarters state, although Compustat firms operate across many states, so multi-state employee exposure remains unresolved and would tend to attenuate the estimated effects toward zero.

This extension's main contribution is a framework that connects state economic freedom and firm size as theoretically related moderators of PFL's effect on firm risk-taking: through managerial-discretion theory they are external and internal conditions shaping the latitude, and the resources, with which firms convert the PFL threat response into action, and their interaction offers a directional way to compare a discretion-amplification with a financial-constraints account. Worker retention is left as the natural next step for future work: PFL is known to reduce worker turnover (Bennett, Erel, Stern, and Wang, 2025), and whether part of the risk-taking response operates through a more stable workforce is an exploratory channel to test once firm-level turnover data and additional treated cohorts make a credible mediation design feasible. Empirically, what the pilot supports most clearly is the firm-size heterogeneity: the PFL effect is concentrated among larger firms and is essentially absent for SMEs, a pattern that is consistent with Lin's size

moderation though not a strict replication, given the different sample and measurement. The economic-freedom moderation and the firm-size-by-freedom connection are weaker and only directional. The paper therefore offers a promising framework together with a directional pilot, with the size result as its firmest empirical contribution and the economic-freedom story as a hypothesis for future, better-powered tests as additional treated cohorts mature.

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Appendix. Rhode Island pre-trend diagnostic

Lin (2025) excludes Rhode Island from the main analysis for failing a parallel-trends check; Figure A1 reproduces that diagnostic to confirm the reasoning. Rhode Island tracks the never-treated control mean through roughly 2005 but then rises sharply, years before its 2014 PFL effective date, violating the parallel-trends assumption. The cohort is also very small (4 firms, 51 firm-years), so the figure is descriptive rather than a powered test. As reported in Section 6.5, the main results are robust to including Rhode Island, so the exclusion sharpens identification without driving the findings.

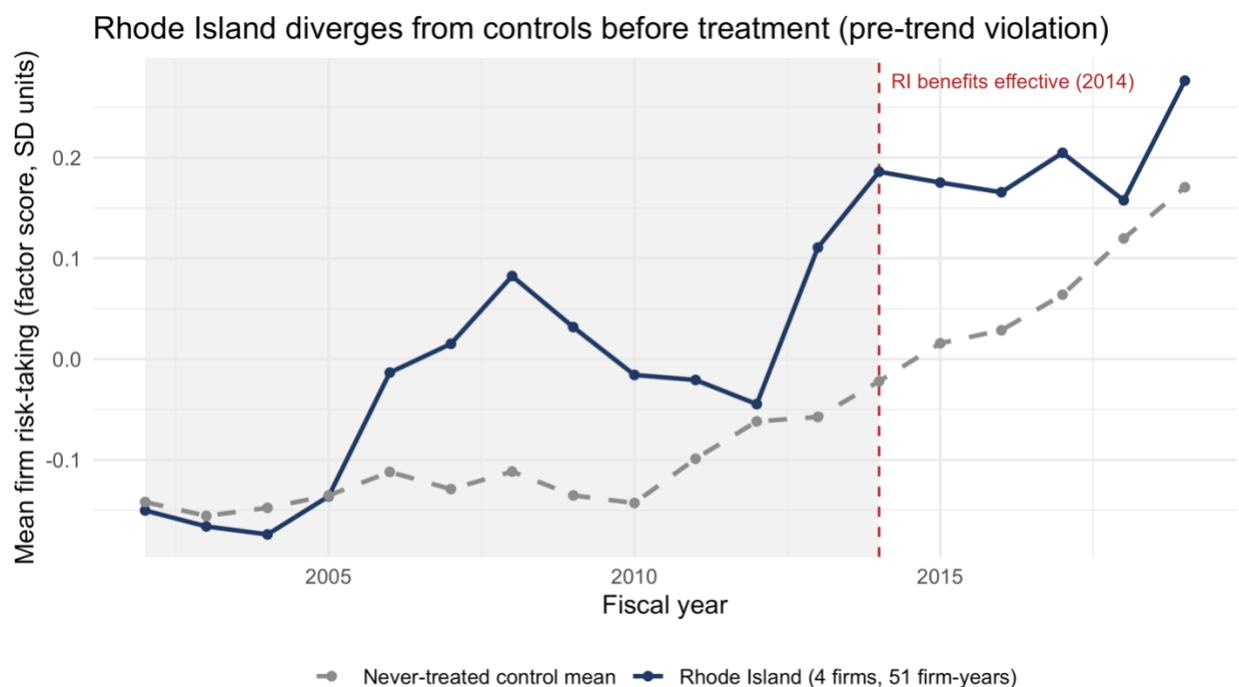


Figure A1. Rhode Island versus never-treated control mean firm risk-taking (factor score, SD units), 2002-2019. The dashed line marks Rhode Island's 2014 PFL effective date; Rhode Island diverges upward years earlier, before treatment.